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OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 1

Team No.: 4

Project Title: Linux Dynamic Memory Allocation and Memory Monitoring System

Team Members

Roll Number	Student Name	Signature
2520030601	Billakanti Siri	
2520030082	Bhavana Kondakrindi	

1. Abstract

Memory management is an important function of an operating system, as inefficient allocation and deallocation can lead to memory fragmentation, excessive memory usage, and memory leaks. This project proposes a Linux-based **Dynamic Memory Allocation and Memory Monitoring System** to provide a practical understanding of how memory is allocated, released, and monitored during program execution.

The system will allow users to perform dynamic memory allocation and deallocation while maintaining a record of active memory blocks. It will monitor memory usage, track allocation and deallocation operations, identify possible memory leaks, and analyze memory fragmentation. Different memory allocation strategies such as First Fit, Best Fit, and Worst Fit can also be implemented and compared based on memory utilization and allocation efficiency.

The project will be developed using C/C++ on Linux and will utilize memory-management mechanisms and system-level interfaces such as malloc(), free(), realloc(), mmap(), munmap(), and Linux /proc information where applicable. The system will generate memory usage statistics and analysis reports to help users understand memory-management behavior.

The expected outcome is a functional Linux-based tool that demonstrates dynamic memory management, provides useful memory monitoring and analysis, and strengthens practical understanding of Operating Systems and Linux system programming concepts.

2. Problem Statement

Efficient memory management is essential for the proper functioning of an operating system. During program execution, frequent memory allocation and deallocation can result in inefficient memory utilization, memory fragmentation, and memory leaks. However, it is difficult for users to observe and analyze these memory-management activities directly.

The proposed project addresses this problem by developing a Linux-based system that tracks dynamic memory allocation and deallocation, monitors memory usage, and analyzes memory-management behavior. The system will provide information about active memory blocks, memory utilization, fragmentation, and possible memory leaks. It will also support the comparison of different memory allocation strategies to study their efficiency.

The project is relevant to Operating Systems and Systems Programming because it demonstrates practical concepts of dynamic memory management, memory allocation, process memory monitoring, and interaction with Linux memory-management mechanisms and system-level APIs.

3. Objectives

1. To implement dynamic memory allocation and deallocation in a Linux environment.
2. To track and monitor memory blocks allocated and released during program execution.
3. To measure memory utilization and identify memory fragmentation and possible memory leaks.
4. To implement and compare First Fit, Best Fit, and Worst Fit memory allocation strategies based on their efficiency.
5. To utilize Linux memory-management mechanisms and system-level APIs to analyze memory behavior.
6. To generate clear memory usage statistics and reports for analyzing the performance of different allocation strategies.

4. Proposed Methodology

The proposed system will be developed as a Linux-based application using C/C++. It will maintain a record of memory blocks requested and released during program execution. The system will provide options to allocate, deallocate, and reallocate memory and will track the size and status of each memory block.

First, the system will receive memory allocation requests and apply selected allocation strategies such as First Fit, Best Fit, and Worst Fit. The allocated and free memory blocks will be maintained and updated

after every operation. The system will continuously collect memory usage information and analyze the available and utilized memory.

The system will further analyze the allocation records to identify memory fragmentation and possible memory leaks. Linux memory-management mechanisms and system-level interfaces will be used where applicable to obtain memory-related information. Finally, the collected data will be processed to generate memory usage statistics, allocation performance comparisons, and analysis reports.

The overall flow will be:

Memory Request → Allocation Strategy → Memory Allocation → Memory Monitoring → Deallocation → Fragmentation/Leak Analysis → Report Generation

5. Operating Systems Concepts / Linux APIs Used

OS Concept / Linux API / System Call	Purpose in the Project
Dynamic Memory Allocation	Allocate and release memory blocks during program execution.
malloc()	Dynamically allocate a specified amount of memory.
free()	Release previously allocated memory and make it available for reuse.
realloc()	Resize an existing dynamically allocated memory block.
mmap()	Create memory mappings for studying and managing memory at a lower system level.
munmap()	Release memory mappings created using mmap().
/proc Filesystem	Obtain Linux process and memory-related information for monitoring and analysis.
Memory Fragmentation	Analyze how allocation and deallocation operations affect available memory blocks.

6. Individual Contribution

Roll Number	Student Name	Individual Responsibility
2520030601	Billakanti Siri	Design and implement the dynamic memory allocation module, including malloc(), free(), and realloc() operations. Implement First Fit, Best Fit, and Worst Fit allocation strategies. Develop memory allocation tracking and fragmentation analysis.

2520030082	Bhavana Kondakrindi	Design and implement the memory monitoring and analysis module. Develop memory usage tracking, memory leak detection, Linux /proc-based monitoring, and memory statistics/report generation. Handle testing and performance analysis of the system.

7. Tools / Platforms / Software Used

Tool / Platform / Software	Purpose
Linux / Ubuntu	Operating environment for developing and executing the Linux-based memory management system.
C / C++	Primary programming language for implementing memory allocation, monitoring, and analysis modules.
GCC (GNU Compiler Collection)	Compiling and building the C/C++ source code.
GDB (GNU Debugger)	Debugging the program and identifying memory-related issues during development.
Git & GitHub	Version control, source code management, and project repository hosting.

8. Expected Outcome

The project is expected to produce a functional Linux-based memory management and monitoring system that can perform dynamic memory allocation, deallocation, and reallocation while maintaining records of memory usage. The system will provide memory utilization statistics and identify memory fragmentation and possible memory leaks.

It will also demonstrate and compare First Fit, Best Fit, and Worst Fit allocation strategies based on memory utilization and allocation efficiency. The system will generate clear reports of allocation activities, memory usage, fragmentation, and detected memory leaks.

The expected outcome is a practical Linux system programming application that helps users understand and analyze dynamic memory management through real-time tracking, performance comparison, and system-level memory monitoring.

9. GitHub Repository

GitHub Repository Link: https://github.com/siribillakanti/TechTitans_OSSP

https://github.com/bhavanakondakrindi/TechTitans_OSSP

Repository Created: _____

□ Yes □ No

Faculty Name and Signature

Faculty Name: _____

Signature: _____

Remarks: _____

Faculty Use

Project Approved: _____

□ Yes □ No

Date: _____

Faculty Signature: _____